

Stage 11 Revision-43 Dependency, Temperature, and Quality Consistency Specification

2026-07-27

Purpose

This specification owns the corrections through architecture revision 43. It is a documentation and planning contract; it does not change existing S0–S4 runtime algorithms or scientific outputs.

Required corrections

- E0b owns immutable raw availability only; admissibility is attached by signed overlays referencing ENS/STAT certificates.
- No untested stationarity assertion may promote PMF or E5 evidence.
- Source energy channels are reconstructed before conservation analysis.
- STAT1 production-regime selection uses fixed source-level observables; held-out site-density stability is a later STAT3 refinement.
- Candidate discovery and basin/corridor validation are cross-fitted across disjoint complete-system blocks.
- E2 numerical density boundaries are not called validated saddles.
- NVT, NVE, NpT, and biased thermodynamics use distinct potential definitions.
- MBAR requires cross-evaluated reduced potentials or a certified equivalent map.
- Tagged-ion, pooled-orbit, occupation-vector, and joint-state probabilities are distinguished explicitly.
- Formal zero-event rate bounds require an E9/11F `RateBoundModel`.
- Source-control architecture is generic, with VASP as a versioned adapter.
- Current normative stage order appears once; descriptive history is maintained in a separate appendix.

Acceptance

The normative manual must contain revision 43, `SimulationRunControls`, `ProductionRegimeCatalog`, `EvidenceAdmissibilityOverlay`, `11E-STAT3`, the cross-fitted SAMP sequence, ensemble-specific thermodynamic sections, and the reduced-potential-matrix requirement. It must not contain an untested-stationarity promotion path, a current “next implementation boundary” statement, or a source-authoritative ensemble or temperature claim derived from the legacy pilot label.

Revision-43 additions

- Ionic temperature is reconstructed from ionic kinetic energy and a signed active degree-of-freedom count using equipartition.
- Mean, standard deviation, autocorrelation-aware confidence, and drift are distinct reported quantities.
- The VASP adapter extracts numerical MD quality controls and per-step SCF traces from `vasprun.xml`.
- The only execution-quality outcomes are `strictly_qualified`, `degraded_quality`, and `unqualified`.
- Degraded quality warns and proceeds; only unqualified catastrophic integrity failure raises by default.
- Trajectory quality and method-specific thermodynamic admissibility remain separate.